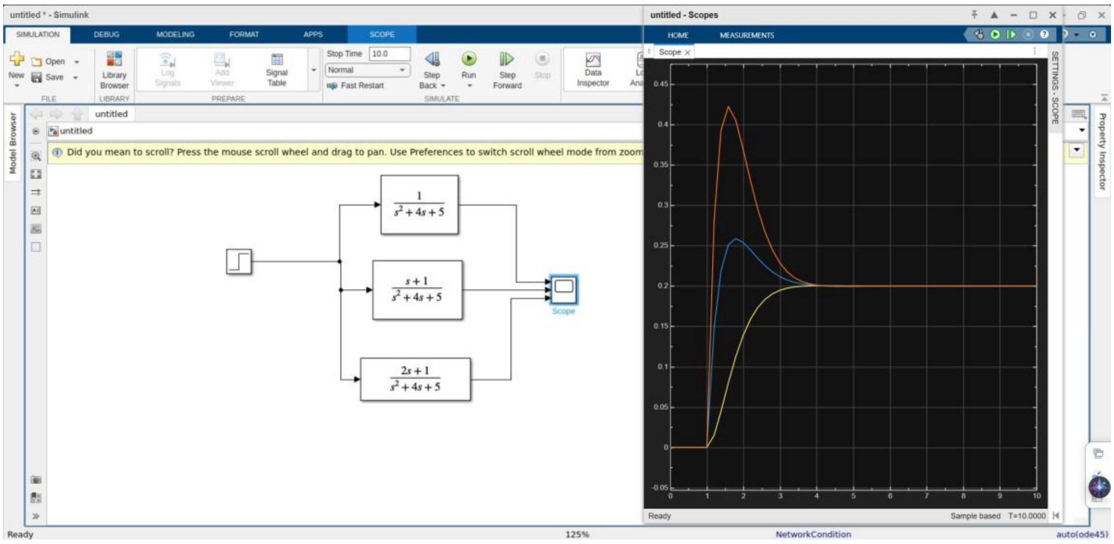
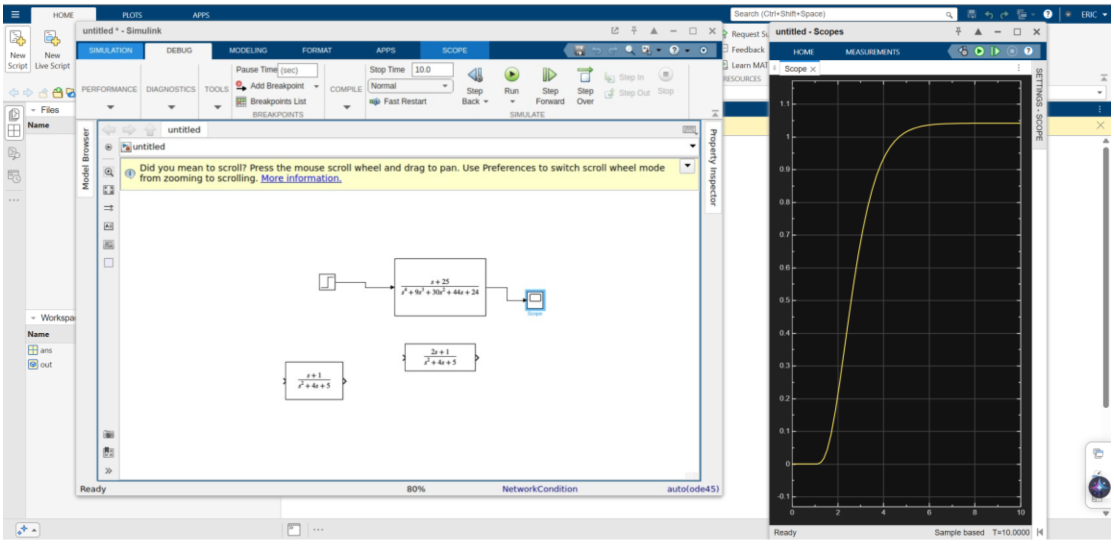
第一題

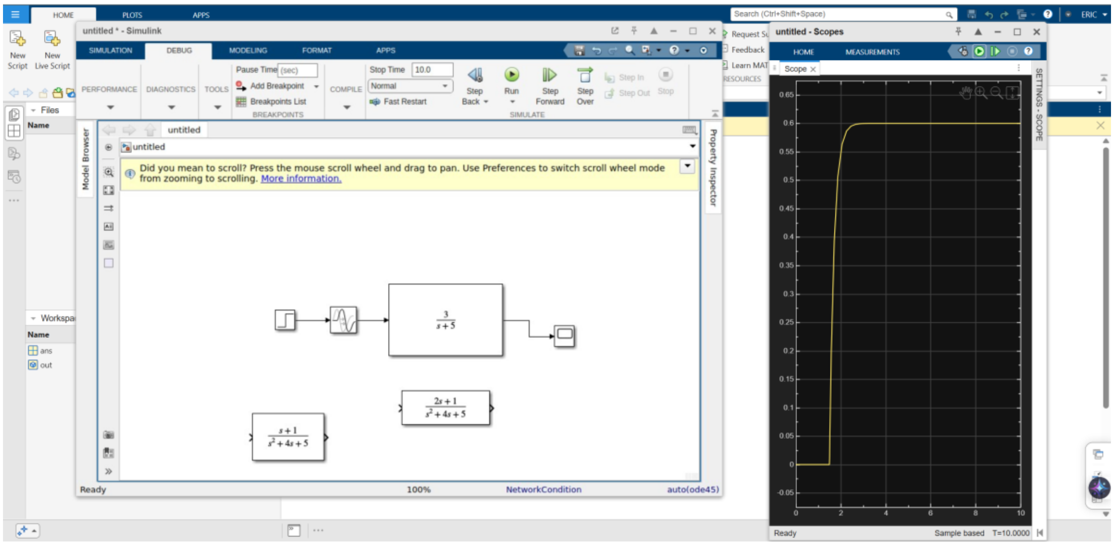
（1）



（2）



（3）



第二題

(1)

```
>> sys1 = tf([1], [1 1])

sys1 =

    1
  -----
  s + 1
```

```
>> sys2 = tf([1 -20], poly([2 2 5]))

sys2 =

             s - 20
  ------------------------
  s^3 - 9 s^2 + 24 s - 20
```

(2)

```
>> sysp = parallel(sys1, sys2)

sysp =

       s^3 - 8 s^2 + 5 s - 40
  -----------------------------
  s^4 - 8 s^3 + 15 s^2 + 4 s - 20
```

(3)

```
>> syss = series(sys1, syss2)
```

(4)

```
>> sysf = feedback(sys1, 1)

sysf =

    1
  -----
  s + 2
```

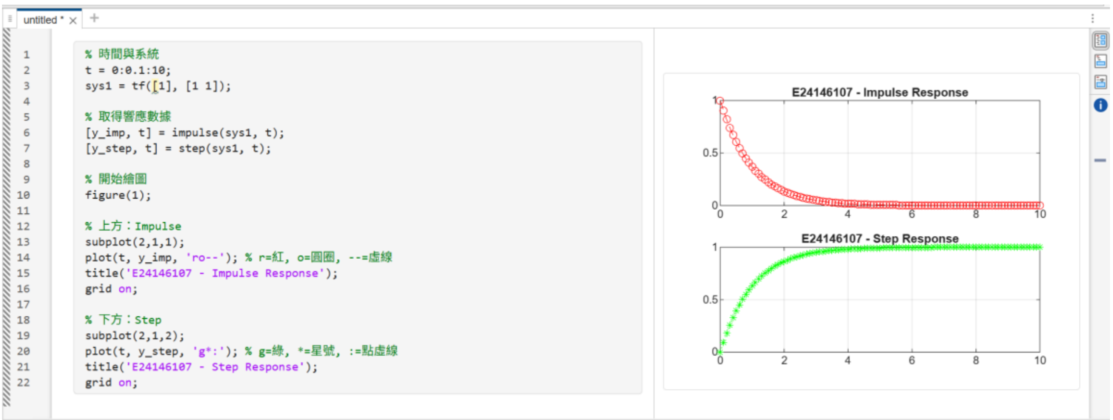
(5)

```
>> [num_all, den_all] = tfdata(sysf, 'v')

num_all =

     0     1


den_all =

     1     2



>> [z_all, p_all, k_all] = zpkdata(sysf, 'v')

z_all =

  0×1 empty double column vector


p_all =

    -2


k_all =

     1
```

```
>> get(sysf)
       Numerator: {[0 1]}
     Denominator: {[1 2]}
        Variable: 's'
         IODelay: 0
      InputDelay: 0
     OutputDelay: 0
       InputName: {''}
       InputUnit: {''}
      InputGroup: [1×1 struct]
      OutputName: {''}
      OutputUnit: {''}
     OutputGroup: [1×1 struct]
           Notes: [0×1 string]
        UserData: []
            Name: ''
              Ts: 0
        TimeUnit: 'seconds'
    SamplingGrid: [1×1 struct]
```

加分

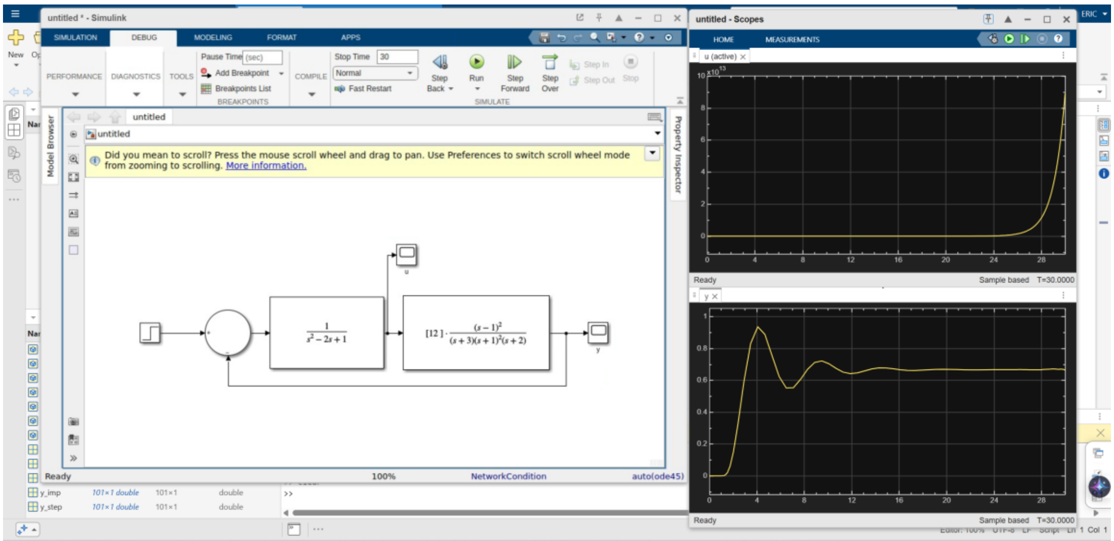
第一題


untitled *
% 時間與系統
t = 0:0.1:10;
sys1 = tf([1], [1 1]);
% 取得響應數據
[y_imp, t] = impulse(sys1, t);
[y_step, t] = step(sys1, t);
% 開始繪圖
figure(1);
% 上方：Impulse
subplot(2,1,1);
plot(t, y_imp, 'ro--'); % r=紅, o=圓圈, --=虛線
title('E24146107 - Impulse Response');
grid on;
% 下方：Step
subplot(2,1,2);
plot(t, y_step, 'g*:'); % g=綠, *=星號, :=點虛線
title('E24146107 - Step Response');
grid on;
E24146107 - Impulse Response
E24146107 - Step Response

第二題

(1)

(2)


untitled * - Simulink
untitled - Scopes
Did you mean to scroll? Press the mouse scroll wheel and drag to pan. Use Preferences to switch scroll wheel mode from zooming to scrolling. More information.
Ready
100%
auto(ode45)